New center of mass: $\frac{1}{9}\left[(\omega_1+1)(0,1)+(\omega_2+1)(8,1)+(\omega_3+1)(2,4)\right]$

$$=\frac{1}{9}\left[((\omega_1+1)\cdot 0,(\omega_1+1)\cdot 1)+((\omega_2+1)\cdot 8,(\omega_2+1)\cdot 1)+((\omega_3+1)\cdot 2,(\omega_3+1)\cdot 4)\right]$$

(Scalar multiplication)

$$=\frac{1}{9}\left[(0,\omega_1+1)+(8\omega_2+8,\omega_2+1)+(2\omega_3+2,4\omega_3+4)\right]$$ (vector addition)

$$=\frac{1}{9}\left[(0+8\omega_2+8+2\omega_3+2,\ \omega_1+1+\omega_2+1+4\omega_3+4)\right]$$

$$=\frac{1}{9}\left[(8\omega_2+2\omega_3+10,\ \omega_1+\omega_2+4\omega_3+6)\right]$$ (Scalar multiplication)

$$=\left(\frac{8\omega_2+2\omega_3+10}{9},\ \frac{\omega_1+\omega_2+4\omega_3+6}{9}\right)$$

Given, $\left(\frac{8\omega_2+2\omega_3+10}{9},\ \frac{\omega_1+\omega_2+4\omega_3+6}{9}\right)=(2,2)$

$8\omega_2+2\omega_3+10=18,\ \omega_1+\omega_2+4\omega_3+6=18$

$8\omega_2+2\omega_3+10=18,\ \omega_1+\omega_2+4\omega_3=12$

$\therefore 8\omega_2+2\omega_3=8,\ \omega_1+\omega_2+4\omega_3=12$

$\boxed{\text{System of equations:}}$

1. $\omega_1+\omega_2+\omega_3=6$
2. $\omega_1+\omega_2+4\omega_3=12$
3. $8\omega_2+2\omega_3=8$

$\boxed{\text{Augmented matrix:}}$

$$\begin{bmatrix}1&1&1&6\\1&1&4&12\\0&8&2&8\end{bmatrix}\xrightarrow[\text{Row2}-\text{Row1}]{\text{Row2}=}\begin{bmatrix}1&1&1&6\\0&0&3&6\\0&8&2&8\end{bmatrix}$$